Preserving Visual Evidence in Compressed Multimodal Reasoning

Working Title

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Problem Statement

SketchThinker-R1's style reward barely shortens traces on its own: raising its weight from 0.05 to 0.1 moves tokens only from 65.2 to 64.3, and the reported parity uses short traces of 182–240 tokens rather than the thousand-token regime in the introduction. This plan does not re-estimate that 64% reduction; it tests a separate failure, in which the training-time judge sees neither the image nor the question, so a concise trace can state correct steps and still omit a visual premise.

Proposed Solution

We will test whether supervision that detects missing visual support improves compressed reasoning beyond style or stated-step correctness rewards. Using Qwen2.5-VL-7B-Instruct and 3,000 verifiable geometry and bar-chart problems, we will split by template family before generating traces or image variants and annotate admissible derivations, accepting valid alternative proofs. A sufficient trace must explicitly state correct required visual premises and valid deductions supporting its returned answer; the image checks stated facts without filling omissions. A judge will learn from supported traces, correct-answer traces with premises deleted, false readings, and redundant-step deletions, with independent human checks on held-out problems. Four reward arms—sufficiency, style, stated-step correctness, and zero auxiliary reward—will share compressed cold-start data, initialization, accuracy/format rewards, and the same measured policy-training accelerator-hours including rollouts and judge scoring. Common budget conditioning and a length penalty will support comparisons at matched achieved reasoning lengths, and accuracy will also be reported by quantile of baseline reasoning length. We will evaluate answer accuracy, human-rated sufficiency, visual-premise retention, end-to-end latency, and correctness on both members of answer-changing image pairs; nuisance edits will test invariance. Separate ablations will remove judge image access and omission-negative training. Three training seeds will show training variation, and problem-family intervals will quantify item-sampling uncertainty. Before reinforcement learning, a pilot puts each omitted sentence back and continues only if the change in answer log probability predicts errors better than length; otherwise the omission-negative target is revised.

Key Research Questions

- Under compression, which necessary visual premises disappear relative to nonvisual steps, and, in a pre-specified subset, which premises used by two later steps disappear, after accounting for valid alternative derivations?
- At matched achieved lengths and training compute, does omission-sensitive supervision improve accuracy and explanation sufficiency beyond style and stated-step correctness rewards?
- Do gains persist on answer-changing image pairs and when judge image access or omission-negative training is removed?

Expected Outcome

The study will establish whether explicit visual-support supervision improves the accuracy–length tradeoff and deliver an independently evaluated compression method, including the conditions where it offers no advantage.